

# Analisi III

2.a. 2025/26

Ing. Fisica

Potli

I. FRAGALA



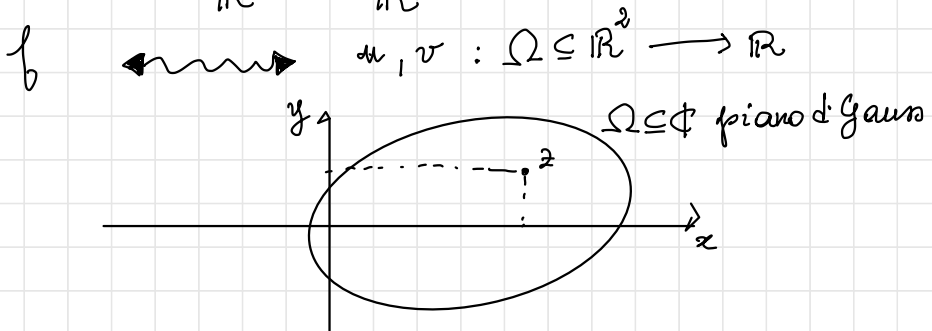
# ANALISI COMPLESSA

Def. Una funzione di variabile complessa

è una funzione  $f: \Omega \subseteq \mathbb{C} \longrightarrow \mathbb{C}$

$$z = x + iy \quad x, y \in \mathbb{R} \quad i^2 = -1$$

$$f(z) = \underbrace{u(z)}_{\uparrow \mathbb{R}} + i \underbrace{v(z)}_{\uparrow \mathbb{R}} = u(x, y) + i v(x, y)$$



$$(u, v) : \Omega \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

$$f: X \longrightarrow Y$$

$$\text{Gr } f = \{ (x, f(x)) \} \subseteq X \times Y$$



- $f(z) = \frac{P(z)}{Q(z)}$ 
 $P, Q$  funzioni polinomiali  
 $\Omega = \{z \in \mathbb{C} : Q(z) \neq 0\}$

$$f(z) = \frac{z^2 - 1}{(z^2 + 1)^2} \quad \Omega = \mathbb{C} \setminus \{\pm i\}$$

- $f(z) = e^z, \sin z, \cos z, \sinh z, \cosh z,$

Funzione  $f(z) = e^z$   $\Omega = \mathbb{C}$

$$z = x + iy \Rightarrow e^z = e^{x+iy} := e^x \cdot \overbrace{(e^{iy})}^{e^{iy}}$$

$$= \underbrace{e^x \cos y}_{u(x,y)} + i \underbrace{e^x \sin y}_{v(x,y)}$$

•  $y=0 \Rightarrow e^z = e^x$

•  $x=0 \Rightarrow e^{iy} = \cos y + i \sin y$

•  $e^{z_1+z_2} = e^{z_1} \cdot e^{z_2}$

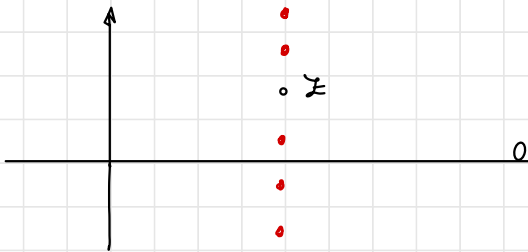
$$z_1 = x_1 + iy_1$$

$$z_2 = x_2 + iy_2$$

$$\left[ e^{z_1} \cdot e^{z_2} = e^{x_1} (\cos y_1 + i \sin y_1) \cdot e^{x_2} (\cos y_2 + i \sin y_2) \right]$$

$$= e^{x_1+x_2} [\cos(y_1+y_2) + i \sin(y_1+y_2)]$$

•  $f(z) = e^z$  è PERIODICA di periodo  $T = 2\pi i$



•  $e^z \neq 0 \quad \forall z \in \mathbb{C}$

$$|e^z| = \sqrt{u^2(x,y) + v^2(x,y)} = \sqrt{e^{2x} \cos^2 y + e^{2x} \sin^2 y} = e^x$$

$$|e^z| = e^{\operatorname{Re} z} \neq 0$$

# Funzioni circolari

$$\cos z := \frac{1}{2} \left( e^{iz} + e^{-iz} \right)$$

$$\sin z := \frac{1}{2i} \left( e^{iz} - e^{-iz} \right)$$

•  $y=0, x \in \mathbb{R}, z=x \Rightarrow$

$$\begin{aligned} \cos z &= \frac{1}{2} \left( e^{ix} + e^{-ix} \right) = \frac{1}{2} \left( \cos x + i \sin x + \cos(-x) + i \sin(-x) \right) \\ &= \frac{1}{2} \cdot 2 \cos x = \cos x \end{aligned}$$

•  $\cos(z + 2\pi) = \cos z$  PERIODICA DI PERIODO  $2\pi$



•  $\cos(x+iy) = \frac{1}{2} \left( e^{i(x+iy)} + e^{-i(x+iy)} \right)$

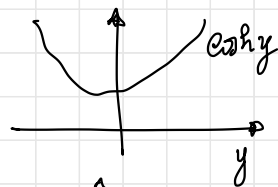
$$= \frac{1}{2} \left( e^{ix-y} + e^{-ix+y} \right)$$

$$\stackrel{\text{def di } e^{\pm i\theta}}{=} \frac{1}{2} \left( e^{-y} (\cos x + i \sin x) + e^y (\cos x - i \sin x) \right)$$

$$= \cos x \cdot \frac{e^y + e^{-y}}{2} - i \sin x \cdot \frac{e^y - e^{-y}}{2}$$

$$= \underbrace{\cos x \cdot \cosh y}_{u(x,y)} - i \underbrace{\sin x \sinh y}_{v(x,y)}$$

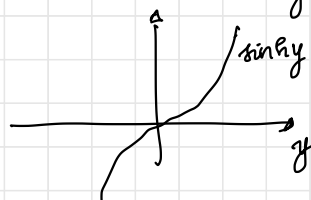
$$\cos z = 0 \iff \begin{cases} \cos x \cosh y = 0 \\ \sin x \sinh y = 0 \end{cases}$$



$$1^{\text{a}} \text{ eq.} \Rightarrow x = \frac{\pi}{2} + k\pi \quad k \in \mathbb{Z}$$

$$2^{\text{a}} \text{ eq.} \Rightarrow \sinh y = 0 \Rightarrow y = 0$$

$$\begin{cases} x = \frac{\pi}{2} + k\pi \\ y = 0 \end{cases}$$



$$f(z) = \sin z \Rightarrow \begin{aligned} u(x, y) &= \sin x \cosh y \\ v(x, y) &= \cos x \sinh y \end{aligned}$$

$$\begin{aligned} \text{oss. } \cos(iy) &= \frac{1}{2} \left( e^{i(iy)} + e^{-i(iy)} \right) \\ &= \frac{1}{2} \left( e^{-y} + e^y \right) = \cosh y \end{aligned}$$

$$\forall y \in \mathbb{R} \quad \boxed{\cos(iy) = \cosh y}$$

$$\bullet \quad \cos^2 z + \sin^2 z = 1 \quad \forall z \in \mathbb{C}$$

Funzioni iperboliche (periodiche di periodo  $T = 2\pi i$ )

$$\cosh z = \frac{e^z + e^{-z}}{2}, \quad \sinh z = \frac{e^z - e^{-z}}{2}$$

$$\cosh z = \cosh x \cosh y + i \sinh x \sinh y$$

$$\sinh z = \sinh x \cosh y + i \cosh x \sinh y$$